

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)			Indicative content: AB and CD have a gradient of 0 which means the acceleration is 0. The train moves at constant velocity between AB and is stopped from CD. BC has a [constant] gradient which is double the value of DE so has a [constant] acceleration that is double. However, BC has a negative gradient and DE a positive gradient. BC is a deceleration or negative acceleration, whereas DE shows a positive acceleration. The area under AB is 4 squares, BC 2 squares, CD 0 squares and DE 4 squares. AB and DE have identical areas so the train travels equal distances in both. In comparison BC is $\frac{1}{2}$ the area of AB or DE so $\frac{1}{2}$ the distance is travelled. 5–6 marks Detailed comparison of distances travelled and accelerations. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3–4 marks Limited comparison of distances travelled and accelerations or detailed comparison of either distances travelled or acceleration. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1–2 marks A brief comparison of either distances travelled or accelerations. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i>	3	3		6	2	

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				0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)		$F = ma$ or accept in words	1			1		
		(ii)	I	Total mass of train = $2.3 \times 10^5 \times 2$ (1) = $[4.6 \times 10^5]$ [kg] = $4.6 \times 10^5 \times 0.7$ = 3.22×10^5 (1) [N] accept 322×10^3 [N] Award maximum of 1 mark for $2.3 \times 10^5 \times 0.7 = 1.61 \times 10^5$ [N] accept 161×10^3 [N] Award 1 mark for answer of 3.22 or 1.61		2		2	2	
			II	Substitution: $0.7 = \frac{55.8}{t}$ (1) Rearrangement: time = 79.7 (1) [s] accept 80 [s]	1	1		2	2	
		(iii)		$2.3 \times 10^5 \times 2 = 4.6 \times 10^5$ (1) which is approximately 4.4×10^5 so this part of claim correct (1) Both trains have same emergency deceleration / 1.2 m/s^2 and so will take the same time to stop / this part of the claim is incorrect (1) Alternative: $\frac{4.4 \times 10^5}{2.3 \times 10^5} = 1.9$ (1) which is approximately 2 so this part of claim correct (1) Both trains have same emergency deceleration / 1.2 m/s^2 and so will take the same time to stop / this part of the claim is incorrect (1)			3	3	1	

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				Alternative: $\frac{4.4 \times 10^5}{2} = 2.2 \times 10^5 \text{ (1)}$ which is approximately 2.3×10^5 so this part of claim correct (1) Both trains have same emergency deceleration / 1.2 m/s^2 and so will take the same time to stop / this part of the claim is incorrect (1)						
				Question 4 total	5	6	3	14	7	0